17th July 2025

POLITECNICO
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A NONPARAMETRIC APPROACH TO INFERENCE AND FORECASTING IN THE ITALIAN ELECTRICITY SPOT MARKET

Nonparametric Statistics
MSc. Mathematical Engineering

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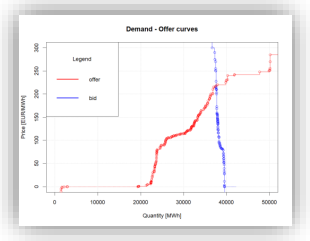
CONTEXT

Platform: **Day-Ahead Market**

Quantity [MWh]
Price [€/MWh]



FINAL PRICE

➢ ANALYSIS OF THE ENVIROMENT:

- Identify **trends** and **tendency** in the Electricity Market
- Identify which **factors** can be *significant* in determining the final price.

➢ PREDICTION:

- **Predict** both curves of Suppy and Demand for the next day, given the observed curves in the previous time instants.

ABOUT THE DATA

Dataset:

Offer – Demand Dataset: provided by Gestore dei Mercati Energetici (**GME**)

Period of interest: **2023 – 2024**

Auction zone : **All the zones**

Data Features:

Fixing

Day

Hour

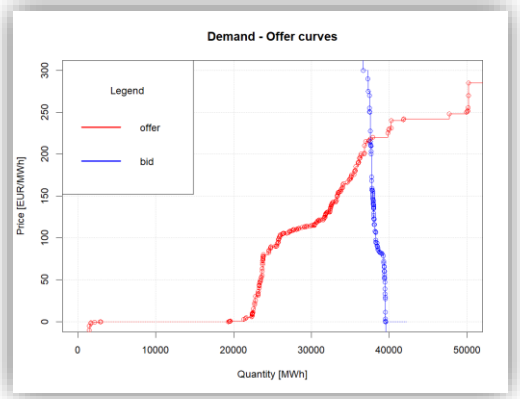
Zone

we obtain a subdataset

Quantity [MWh]

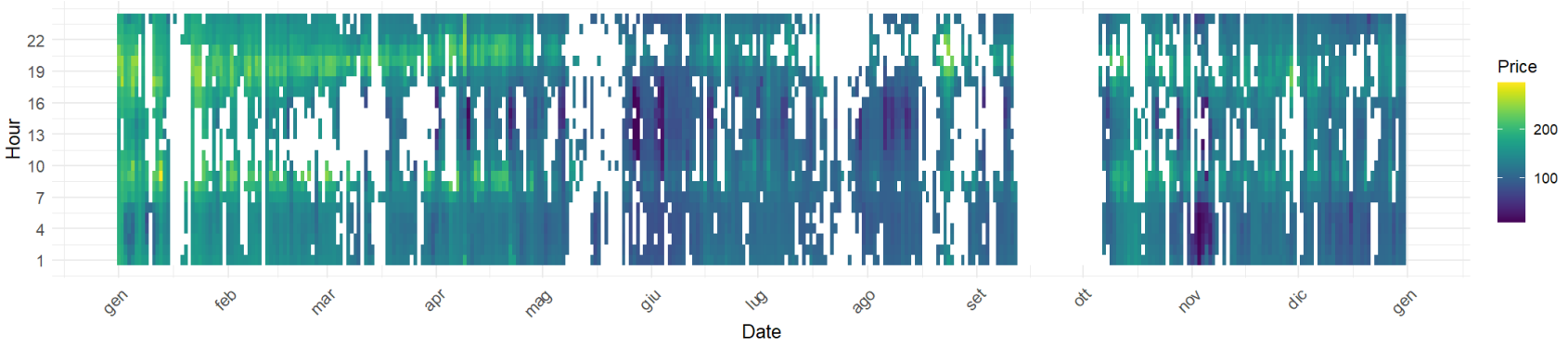
Price [€/MWh]

Type ∈ {OFF, BID}

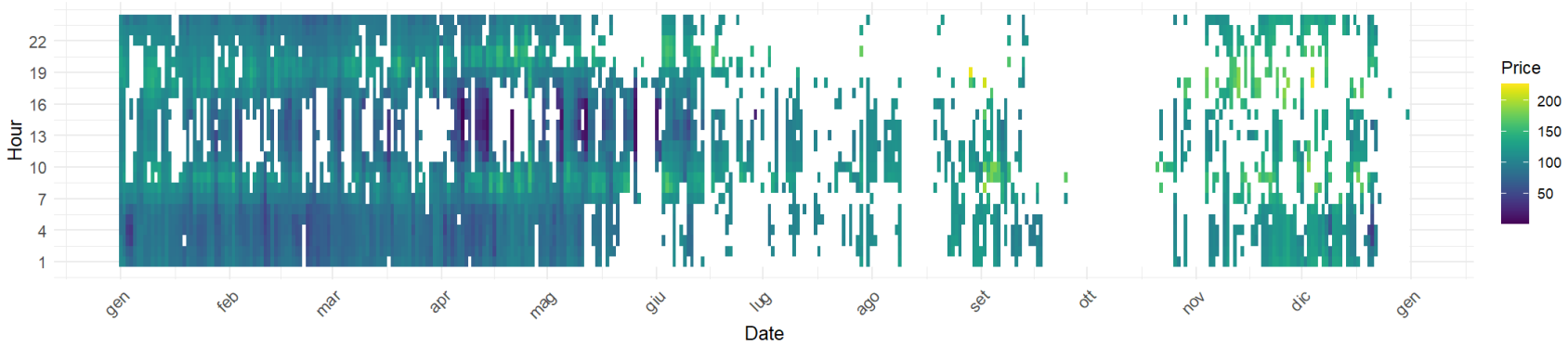


FINAL PRICE EXPLORATION

YEAR 2023



YEAR 2024

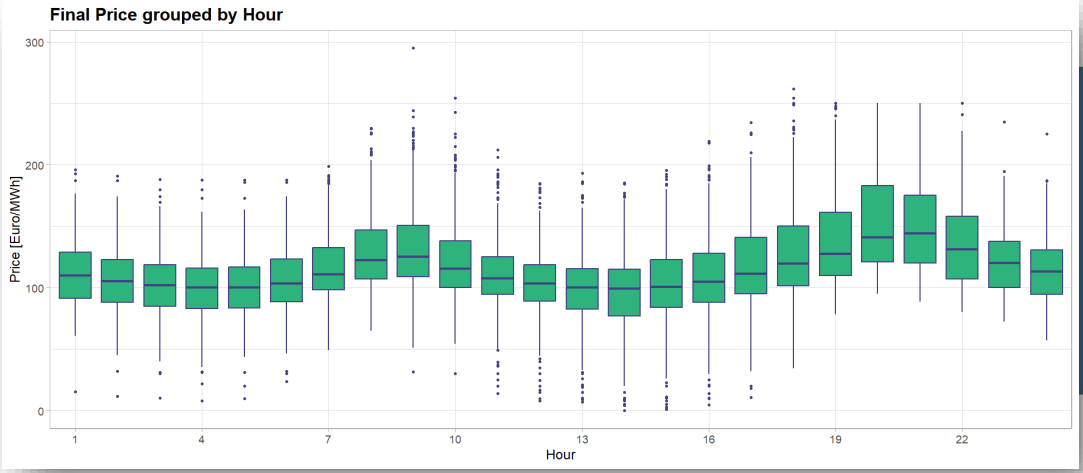


Visual inspection of the final price through heatmaps

- **Regular** and **recurrent pattern** throughout the two-year period
- **Two peaks** typically occurring during the day
- The beginning of 2023 is characterized by **higher prices**
- 2024 data are more **sparse**

TENDENCY DETECTION

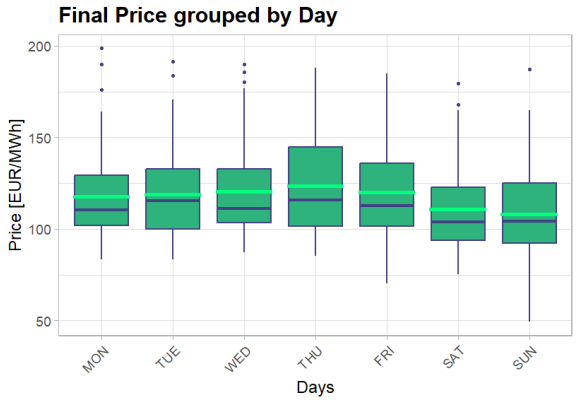
HOUR

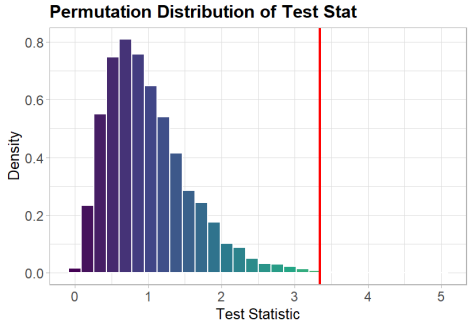


- **Two peaks** around 9:00 AM and 8:00 PM
- More insight about the **distribution**

Consider as baseline for the analysis the obserbations at **7:00 AM**

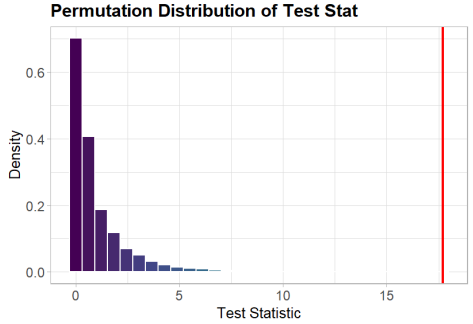
DAYS OF THE WEEK





p-value = **0.003**

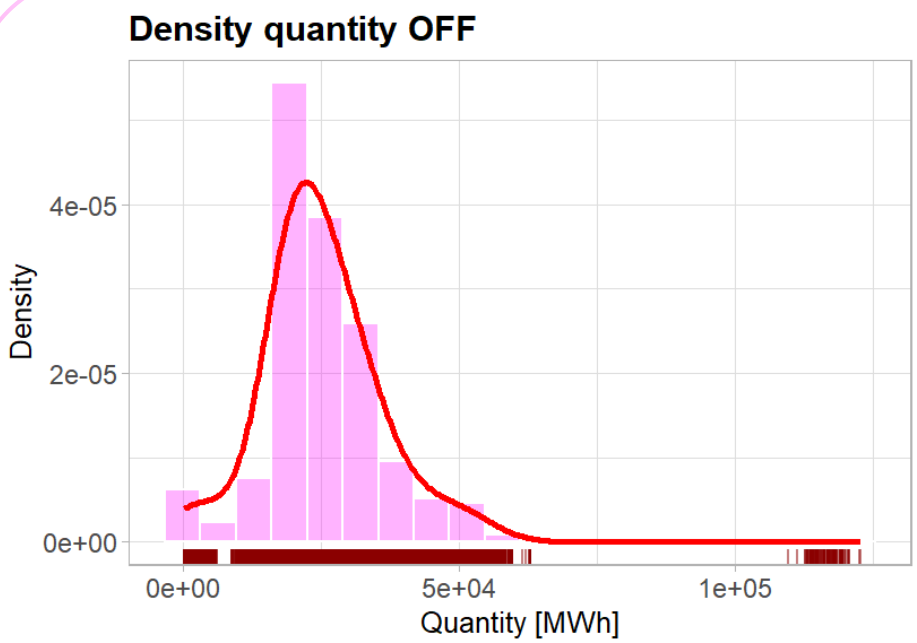
WEEKDAYS vs WEEKEND



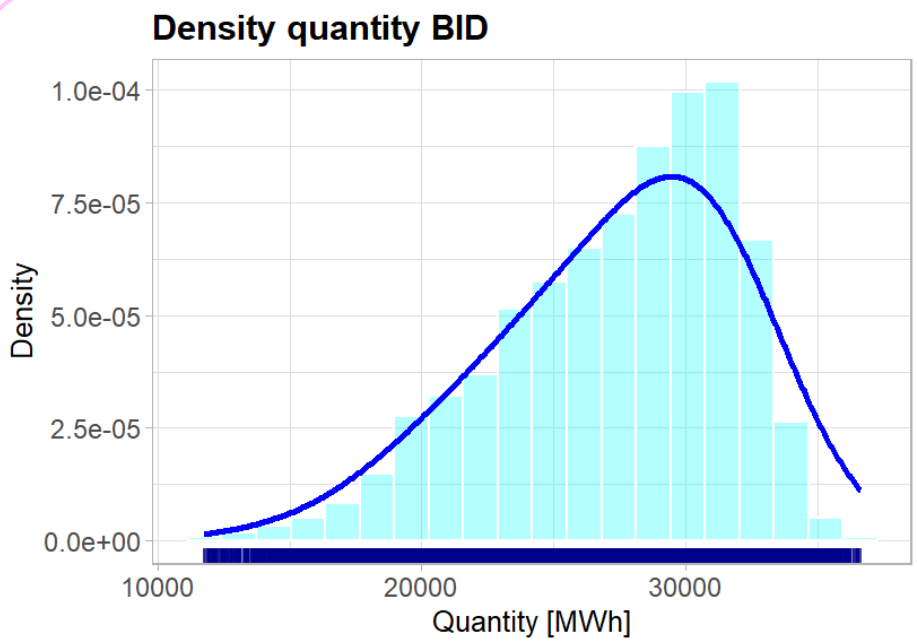
p-value = **0.0**

TOWARD FUNCTIONAL DATA ANALYSIS

Histograms of the bids: which part of the domain needs more complexity



Bandwidth: **5000MWh**



Bandwidth: **2500MWh**

- Range of interest: **[0 – 50.000 MWh]**
- More complexity in the **central bands**

SMOOTHING

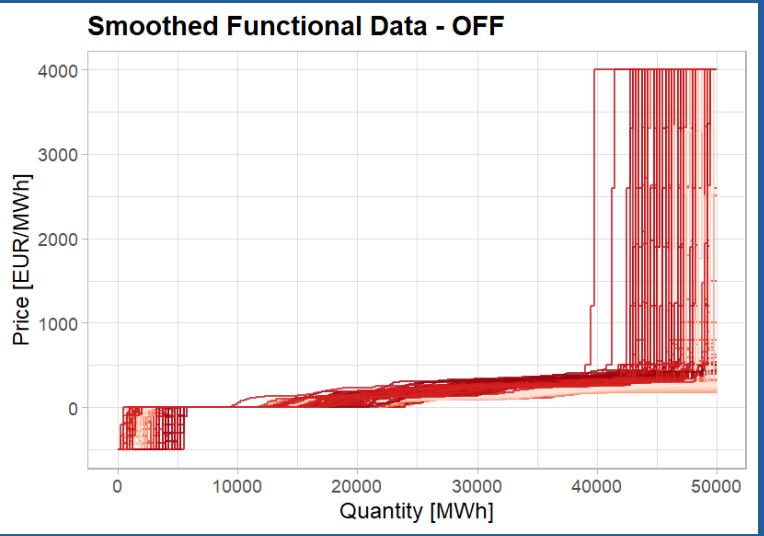
Retrieve proper functional representations of the Supply and Demand curves.
NOTE: the curves are fully observed **step functions**.

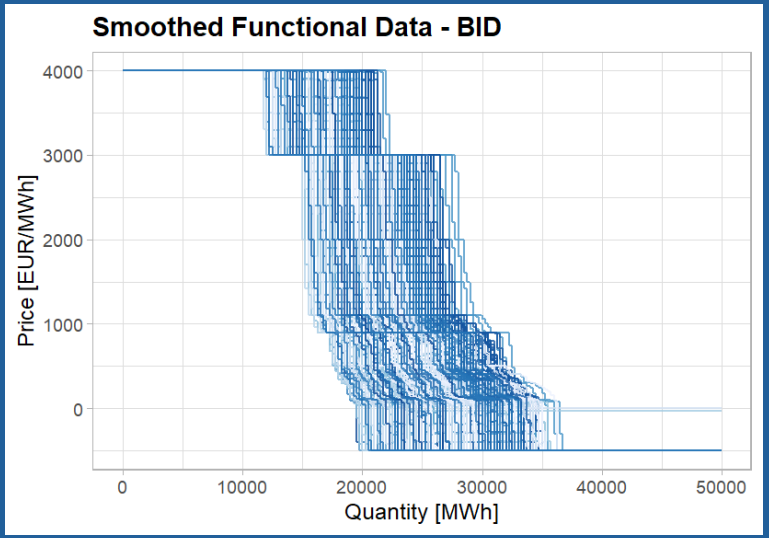
B-spline regression: global method

Used for:
- functional inference
- functional prediction

Supply curve
Knots: 200

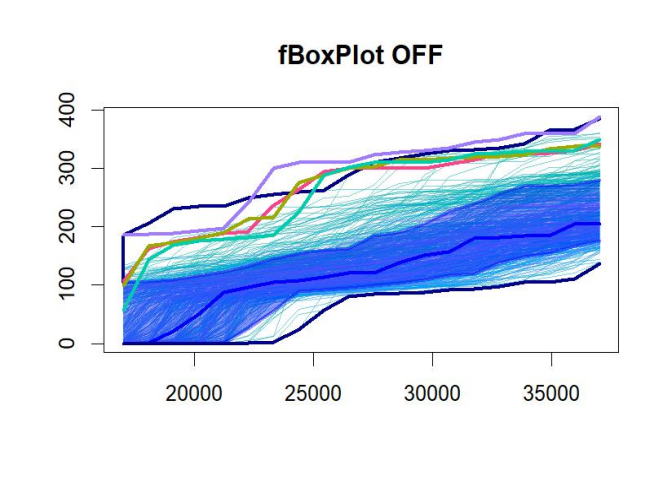
Demand curve
Knots: 200

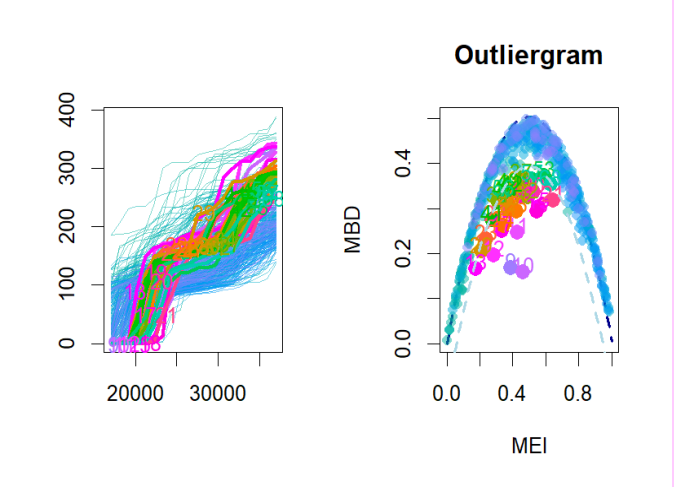




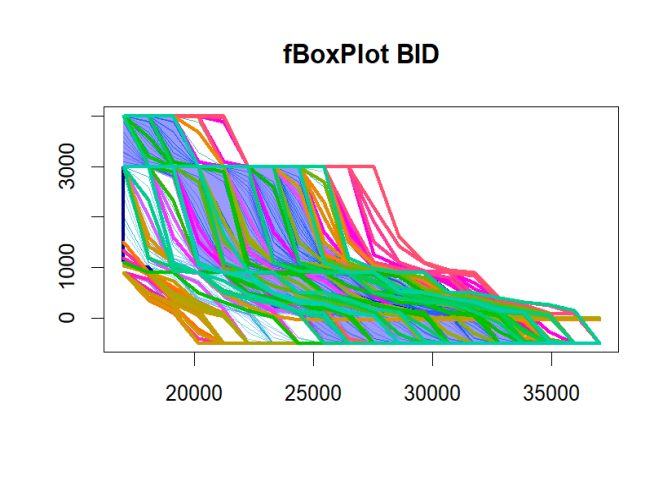
OUTLIER ANALYSIS

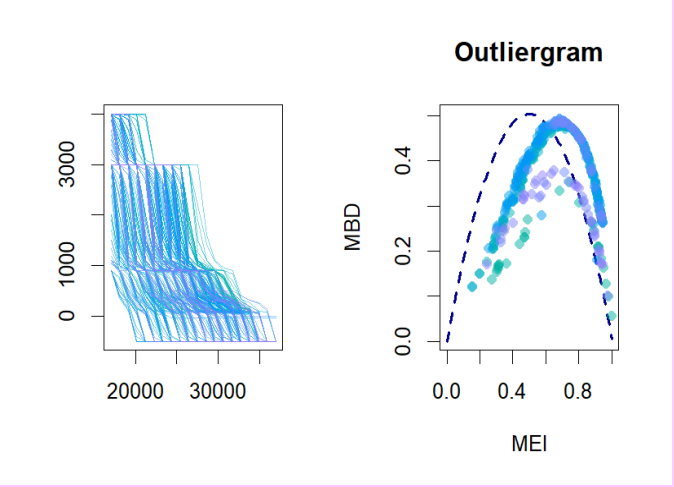
- Few aplitude outliers
- The shape outliers belongs to the **first three month** of 2023





- Many amplitude outliers
- More regular in shape





INFERENCE: TEST WEEKDAYS vs WEEKEND

Functional ANOVA test

Consider two independent (functional) samples

$X_1(t), \dots, X_{n1}(t) \sim iid\ X(t)$
$Y_1(t), \dots, Y_{n2}(t) \sim iid\ Y(t)$

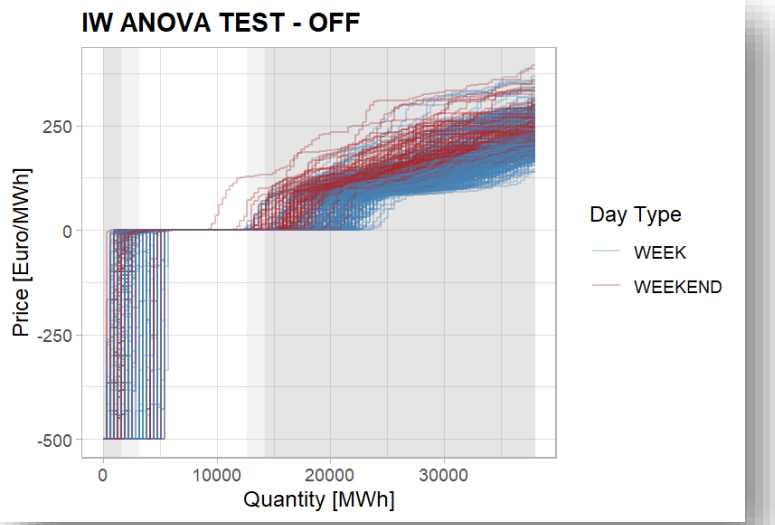
$$H_0: \mathbb{E}[X(t)] = \mathbb{E}[Y(t)] \quad vs \quad H_1: \mathbb{E}[X(t)] \neq \mathbb{E}[Y(t)]$$

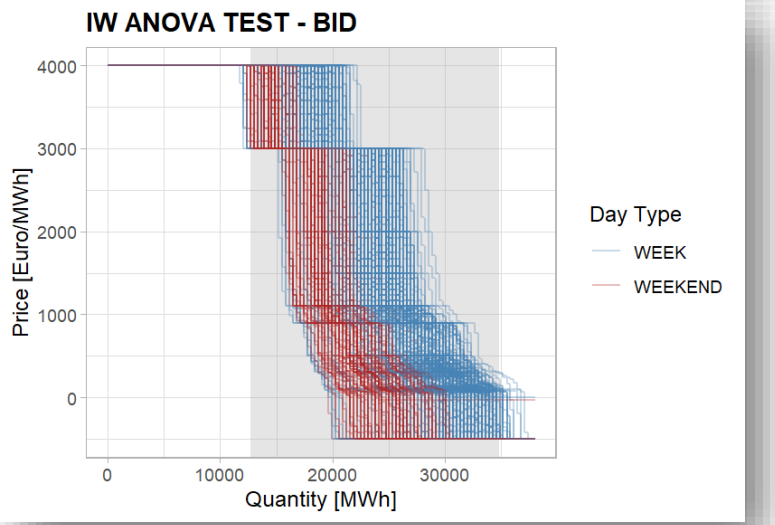
Interval-wise ANOVA test: WEEKDAYS vs WEEKEND

The curves associated with the two groups differ. In particular:

- Upward shift in the supply curves
- Left shift in the demand curves

The shift in the demand curve determine the **average lower price** during the weekend





INFERENCE: TEST HOURS DIFFERENCE

Functional two-sample paired t-test

Consider two paired (functional) samples

$$(X_1(t), Y_1(t)), \dots, (X_n(t), Y_n(t)) \sim iid\ (X(t), Y(t))$$

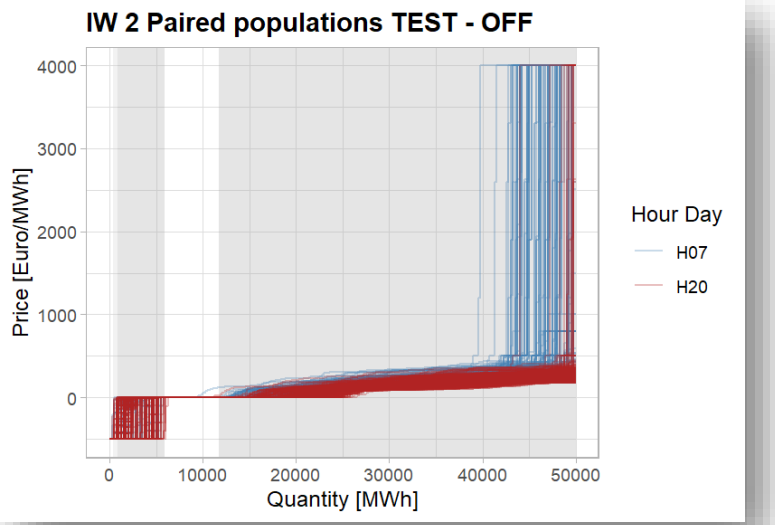
$$H_0: \mathbb{E}[X(t) - Y(t)] = 0 \qquad vs \qquad H_1: \mathbb{E}[X(t) - Y(t)] \neq 0$$

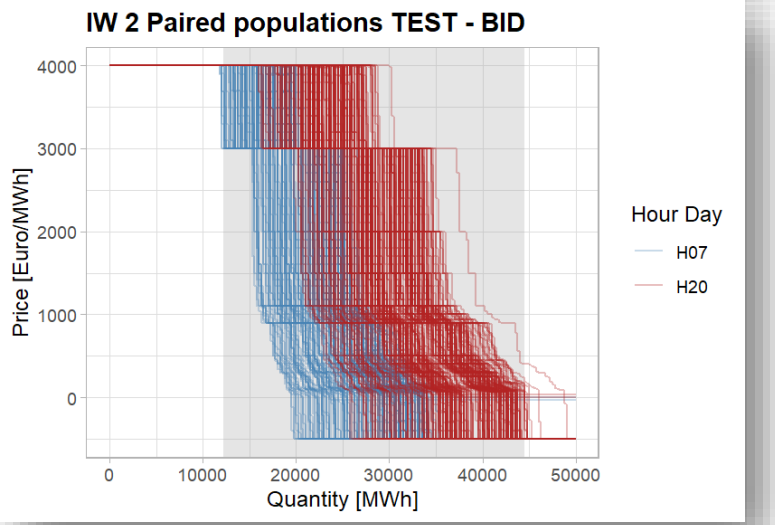
Interval-wise two-sample paired t-test : 7:00 AM vs 8:00 PM

The curves associated with the two groups differ. In particular:

- Downward shift in the supply curves
- Right shift in the demand curves

The shift in the demand curve determine the **average higher price** during 8:00 PM





PREDICTION MODEL: FAR

A **FAR(*p*)**, *functional autoregressive model of order p*, can be defined as:

$$X_n = \alpha + \sum_{i=1}^{p} \Psi_i(X_{n-i}) + \varepsilon_n$$

where in the Hilbertian setting: $\Psi(X)(t) = \int \psi(t,s)X(s)ds$

Assumptions:

$X(t) \in L^2(\mathrm{D})$

$\varepsilon_i \sim iid \quad \mathbb{E}[\varepsilon_i] = 0$

$X(t)$ *stationary*

Consider a **FAR(1):** $X_n = \alpha + \Psi(X_{n-1}) + \varepsilon_t$ where if $\mathbb{E}[X_i(t)] = 0$

$$X_n(t) = \int \psi(t,s)X_{n-1}(s)ds + \varepsilon_n$$

where $\psi(t,s)$ can be estimated using the 1-step lag autocovariance. Morever, exploiting the FPCA:

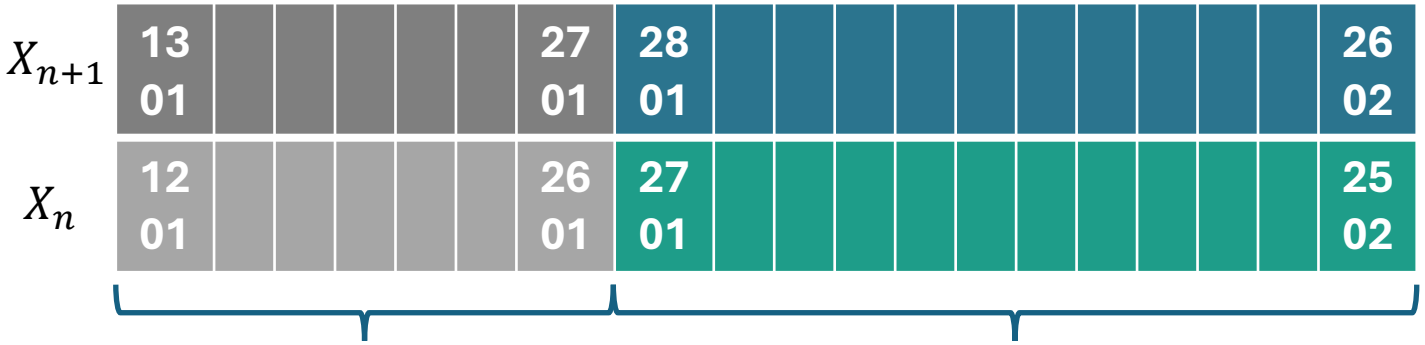
$$\hat{\psi}_{ij} = \hat{\lambda}_i^{-1} \frac{1}{N-1} \sum_{n=1}^{N-1} \langle X_n, \hat{v}_i \rangle \langle X_{n+1}, \hat{v}_j \rangle$$

$$\hat{X}_{n+1}(t) = \sum_{k=1}^{d} \left(\sum_{l=1}^{d} \hat{\psi}_{kl} \langle X_n, \hat{v}_j \rangle \right) \hat{v}_k(t)$$

and finally the mean is added

CONFORMAL PREDICTION INTERVALS

Split of the training set:



Calibration set

Evaluate the non-conformity measure

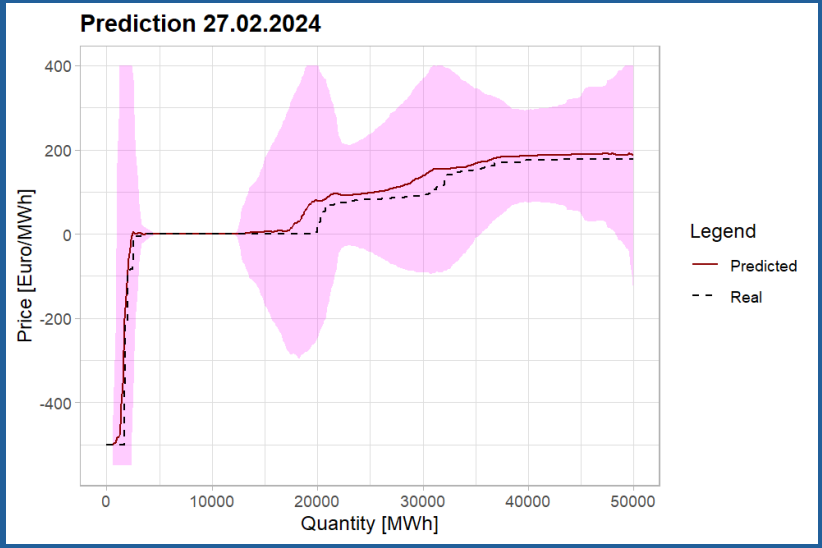
Proper training set

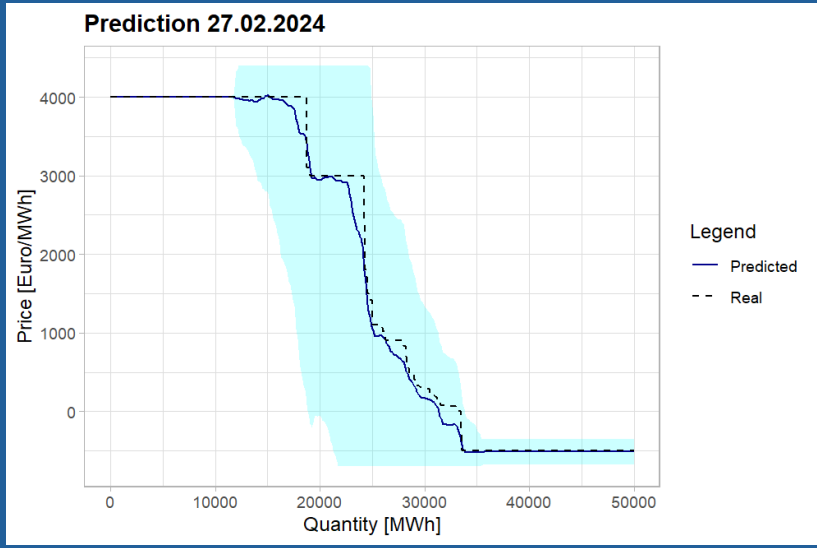
Train the prediction rule i.e. FAR(1)

Non-conformity measure:

$$A(X_{i-1}, X_i) = \sup_{t \in I} \frac{|X_i(t) - \hat{X}_i(t)|}{\sigma(t)}$$

and finally build the CI at level $\alpha = 90\%$ for the prediction $\hat{X}_{n+1}$





CONCLUSIONS

- The study highlights **recurring patterns** in the daily evolution of the final electricity price.
- Daily, **two price peaks** are observed around 9:00 AM and 8:00 PM, driven mainly by **increased demand**. Although supply generally increases during peak hours, this is not sufficient to offset the price rise.
- **Weekend prices are typically lower** due to a leftward shift of the demand curve caused by a **reduced consumption**, despite a more higher supply configuration.
- The **FAR(1)** model was used for forecasting, levereging temporal correlation in functional data. It **provided realistic predictions** even with minimal structure. In contrast, semi-parametric models (e.g. Ridge regression) require more data and tend to produce over-regularized curves. Due to the stationarity assumption, the model's point predictions are centered around the empirical mean.
- **Conformal prediction intervals** were used to quantify the uncertainty, based on the L^∞ norm of residuals **scaled by the local variability**. In the regions with sharp changes, the prediction bands are sometimes too wide, partially due to the limitations of the L^∞ norm as a non-conformity measure.
- The **analysis is purely structural**, relying only on curve shapes and excluding external covariates (e.g. gas prices, renewable generation, inflation), which may be significant in the determination of the final price

REFERENCES

- **Gestore Mercati Energetici**. *Mercato del giorno prima*, http://www.mercatoelettrico.org, 2025.
- **Denis Bosq**. *Linear processes in function spaces: theory and applications*, volume 149. Springer Science&Business Media, 2000
- **Devin Didericksen, Piotr Kokoszka, and Xi Zhang**. *Empirical properties of forecasts with the functional autoregressive model*. Computational statistics, 27(2):285–298, 2012

- The source code is available at: https://github.com/mattiagast/projectNPS